Lecture 4: Nonlinear Forecasting Methods

ECON 417 Business Forecasting

Ivy Yang

Economics & Finance, SIUE Business School

How Lecture 4 Fits the Course

- Lecture 3 stabilized linear models with shrinkage.
- Lecture 4 asks: what if the financial relationship is not linear to begin with?
- Lecture 5 will add uncertainty bands to these richer forecasts.

Key message: Flexibility is useful only if it improves out-of-sample forecasting honestly.

Agenda (Today)

1. Why Nonlinearity Appears
2. Tree-Based Forecasting
3. Feature Importance
4. Decision Rules

Why Linear Models Can Miss Financial Reality

- A small VIX increase during calm markets may have a modest effect on equity returns.
- A sharp volatility spike during a stress episode can produce a disproportionately large negative return.
- Regime thresholds and state-dependent responses are common in financial data.

Practical rule

Start linear, diagnose what it misses, then add flexibility only when the business story and validation results agree.

Polynomial Terms and Interactions

$$r_t^{SP500} = \beta_0 + \beta_1 \Delta \mathrm{VIX}_t + \beta_2 \Delta \mathrm{VIX}_t^2 + \beta_3 \left(\mathbf{1}_{\mathrm{HighVIX},t} \times r_{t-1}^{MSFT} \right) + u_t$$

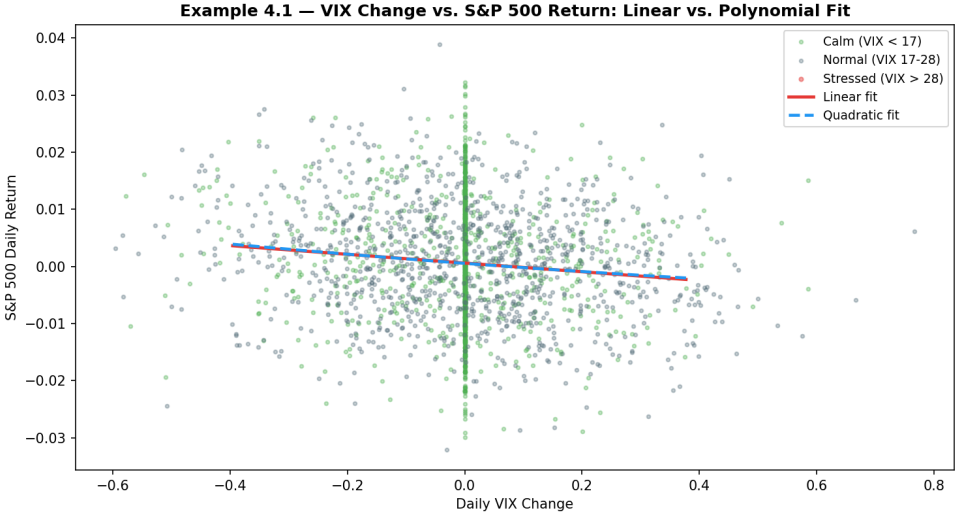
- Polynomial terms add curvature: the return response accelerates as VIX spikes.
- Interaction terms allow the MSFT beta to depend on the volatility regime.
- Forecasting improves when those mechanisms reflect a real market structure.

Key message: "Lagged MSFT predicts SP500" is incomplete. Ask: how much, and *in which regime*?

Notebook Example: VIX Nonlinearity (Example 4.1)

- Same-day VIX changes are plotted against S&P 500 returns (2015–2024).
- A linear fit underestimates large negative moves during volatility spikes.
- A quadratic model captures the accelerating downside curvature.
- Points are color-coded by regime: calm, normal, and stressed (VIX ≥ 25).

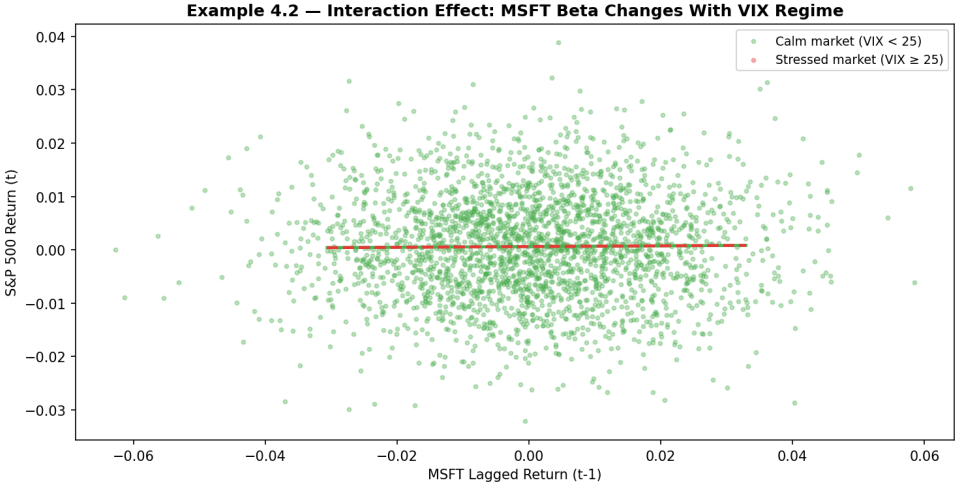
Notebook Example 4.1: VIX–Return Curvature



Notebook Example: VIX Regime $\times$ MSFT Interaction (Example 4.2)

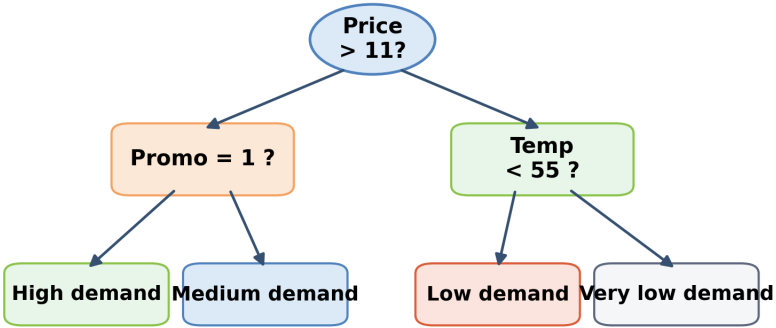
- High-VIX dummy (VIX ≥ 25) is interacted with the lagged MSFT return.
- In calm markets, MSFT's predictive content for SP500 is modest.
- In stressed markets, cross-asset correlations rise: MSFT's beta nearly doubles.
- This pattern—"correlation convergence in crisis"—is well-documented in finance.

Notebook Example 4.2: Calm vs. Stressed Market Betas



Decision Trees

Decision tree = a sequence of simple business rules



What Trees Are Good At

- Trees naturally encode regime thresholds: "if VIX > 20 and SP500 lag < 0, predict negative return."
- They can model interactions without writing them down manually.
- They are easier to explain locally than many other nonlinear methods.

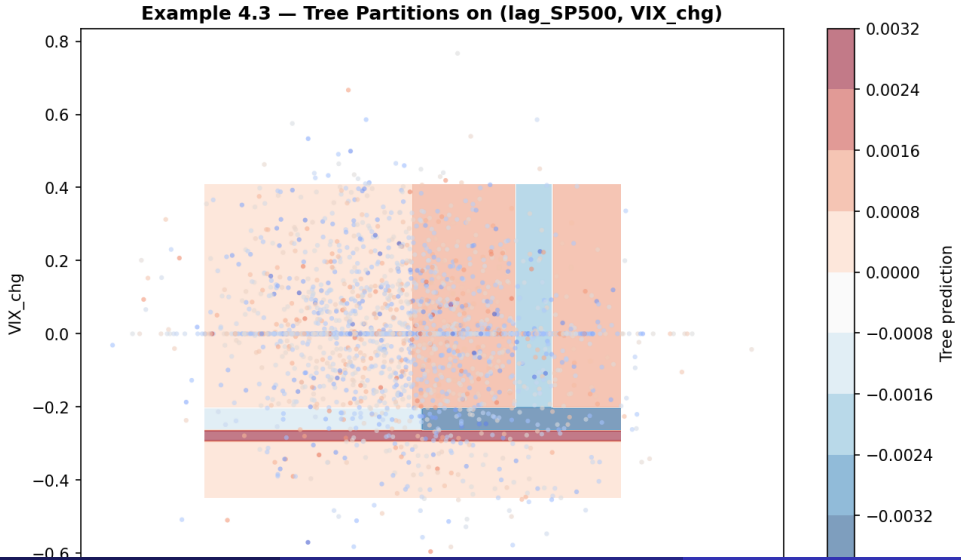
Main risk

A single tree can be unstable: small data changes can create a different sequence of splits.

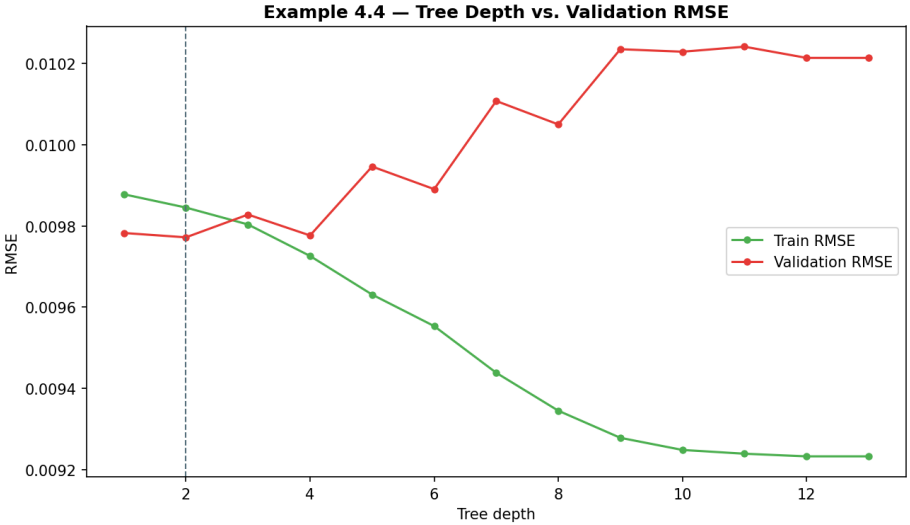
Notebook Example: Tree Partitions (Example 4.3)

- A shallow tree (depth 3) is fitted on lagged SP500 returns and daily VIX changes.
- The rectangular partitions in the 2-D feature space create regime-specific forecast rules.
- Calm-and-up, calm-and-down, stressed-and-up, and stressed-and-down states each receive a different predicted return.

Notebook Example 4.3: Tree Regime Partitions



Tree Depth and Overfitting (Example 4.4)



Random Forest

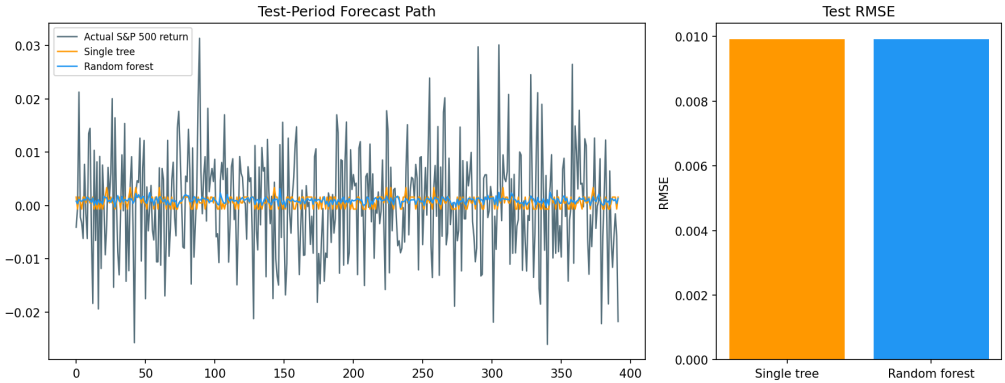
Random forest averages many noisy trees into one stabler forecast

concept adapted from: IBM random forest

Notebook Example: Single Tree vs. Random Forest (Example 4.5)

- The best single tree and a 200-tree forest are compared on the S&P 500 test period.
- The forest produces lower test-set RMSE and a smoother 80-day forecast path.
- Averaging across trees dampens noise from individual splits.

Notebook Example 4.5: Tree vs. Forest on S&P 500 Returns



Feature Importance

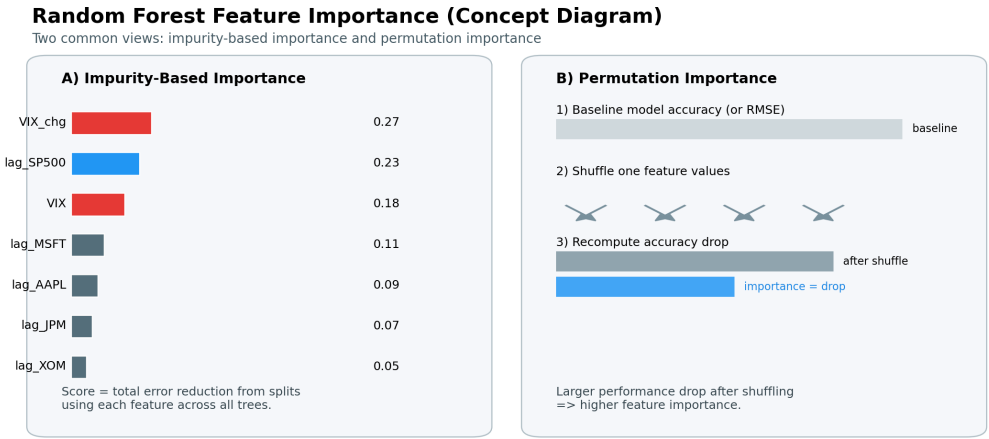
Definition

Feature importance measures how much each predictor reduces forecast error across all trees in the forest, expressed as a fraction of total impurity reduction.

$$\text{Importance}(j) = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \sum_{\text{node } s \in t:\, v(s)=j} \Delta\text{Impurity}(s)$$

- Scores sum to 1 across all features.
- High importance: feature consistently improves splits across many trees.
- Low importance: feature rarely helps and can often be dropped.

Feature Importance: Concept



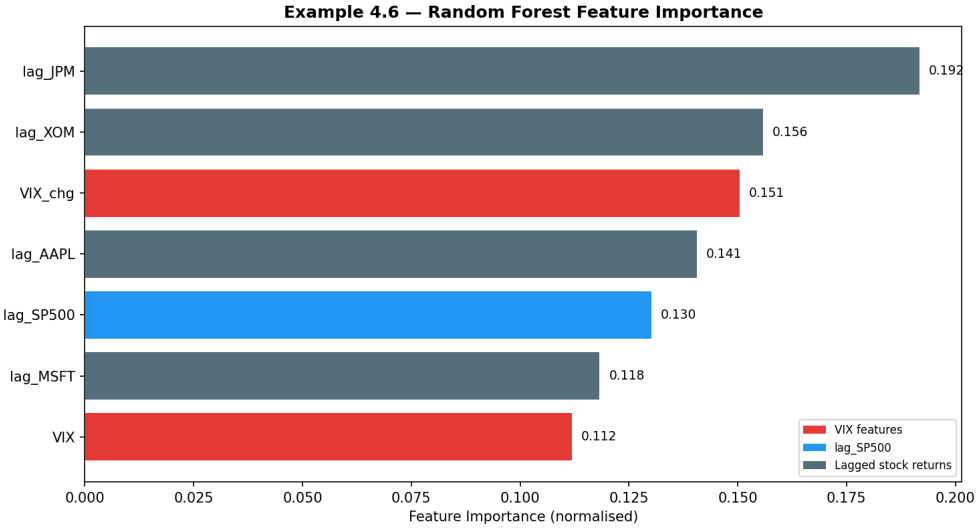
Adapted educational diagram for lecture slides (source theme: IBM Random Forest topic page).

Notebook Example: Feature Importance (Example 4.6)

- Feature importance scores for the S&P 500 return model rank `lag_SP500` and `VIX_chg` as the dominant drivers.
- Lagged individual stock returns (MSFT, AAPL, JPM, XOM) contribute less individually.
- This ranking tells the analyst which signals most deserve monitoring and data-quality investment.

Key message: Importance describes the model, not causal structure. Correlated features share scores.

Notebook Example 4.6: Feature Importance Bar Chart



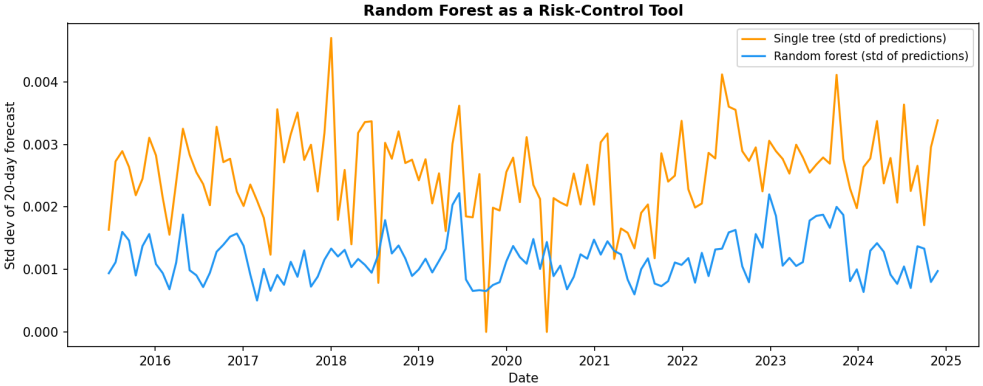
Random Forest as a Risk-Control Tool

Definition: Variance reduction

Averaging predictions from many independent trees reduces forecast variance without assumptions about the functional form of the relationship.

- A single tree can change its forecast substantially after one unusual observation.
- A random forest averages that noise away, producing more stable forecasts over time.
- Stability makes the model easier to defend in risk committees and easier to audit.

Notebook Example: RF Rolling Stability (Example 4.6 companion)



- Rolling prediction variance of single tree vs. forest.
- Forest averaging substantially narrows day-to-day forecast instability.

Single Tree vs. Random Forest Summary

	Single Tree	**Random Forest**
Interpretability	High: readable rules	Lower: average of many
Variance	High: sample-sensitive	Lower: averaging stabilizes
Out-of-sample RMSE	Often weaker	Usually stronger
Governance	Limited	Preferred for risk decisions

Key message: Choose based on validation performance and governance requirements.

When Should We Use Nonlinear Methods?

1. When validation shows the linear model systematically misses curvature or regime interaction.
2. When the nonlinear pattern has a plausible financial explanation (e.g., VIX regime thresholds).
3. When the flexible model beats simple baselines out of sample.
4. When the forecast can still be communicated clearly to a risk committee.

Key message: Nonlinear does not mean "better by default."

Takeaways

- Curvature, regimes, and interactions are common in financial return data.
- Polynomial and interaction terms extend linear regression without changing the estimation framework.
- Trees and forests learn those structures flexibly and rank features automatically.
- Feature importance guides data investment but is not causal attribution.
- Flexible models are more powerful, but they require stricter validation discipline.

Key message: Lecture 5 will add the next practical question: how uncertain are these forecasts?